Assignment 1 Report

Data Mining

NumPy & Pandas Data Preparation and Visualization

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Section: Section 3
Course: Data Mining

Introduction

Overview

In this assignment, I used the **Pima Indians Diabetes** dataset and applied comprehensive data preparation steps using **NumPy** and **Pandas**.

After cleaning and preparing the data, I created **9 visualizations** using **Matplotlib** and **Seaborn** to better understand the dataset, identify patterns, and explore relationships between features and the diabetes outcome.

Dataset Information

Dataset Details

Dataset: Pima Indians Diabetes Dataset
Source: `diabetes.csv`
Rows: 768 **Columns:** 9
Target Column: `Outcome` (0 = No Diabetes, 1 = Diabetes)

Modifications Made in the Dataset

Data Preparation Steps

- Loaded the dataset using `pd.read_csv()` and displayed the first rows with `df.head()` to understand the structure.
- Checked the dataset shape using `df.shape`: confirmed **768 rows** and **9 columns**.
- Displayed column names using `df.columns` and inspected data types with `df.info()`.
- Checked for missing values using `df.isnull().sum()`: found **no NaN values**, but discovered biologically invalid zeros.
- Checked for duplicate rows using `df.duplicated().sum()` and removed any found with `df.drop_duplicates()`.
- Used `df.describe()` to summarize all numerical columns (mean, std, min, max, quartiles).
- Checked class distribution using `df['Outcome'].value_counts()`: **500 non-diabetic** vs **268 diabetic** patients.
- Computed mean values per class using `df.groupby('Outcome').mean()` to compare feature averages between diabetic and non-diabetic groups.
- Replaced biologically invalid zeros in columns `Glucose`, `BloodPressure`, `SkinThickness`, `Insulin`, and `BMI` with the **median** of each column using `df[col].replace(0, np.nan).median()`.
- Removed statistical outliers using the **IQR method**: values below $Q_1 - 1.5 \times IQR$ or above $Q_3 + 1.5 \times IQR$ were dropped for the five cleaned columns.
- Split features and target using `df.drop('Outcome', axis=1)` and `df['Outcome']`.
- Applied `StandardScaler` to normalize all features to zero mean and unit variance, which is essential for SVM performance.
- Split data into training (80%) and testing (20%) sets using `train_test_split()` with `stratify=y` to preserve class proportions.

- Tuned hyperparameters using `GridSearchCV` over a grid of `C`, `gamma`, and `kernel` values with 5-fold `StratifiedKFold` cross-validation to maximize accuracy.

Examples of Specific Changes

Examples

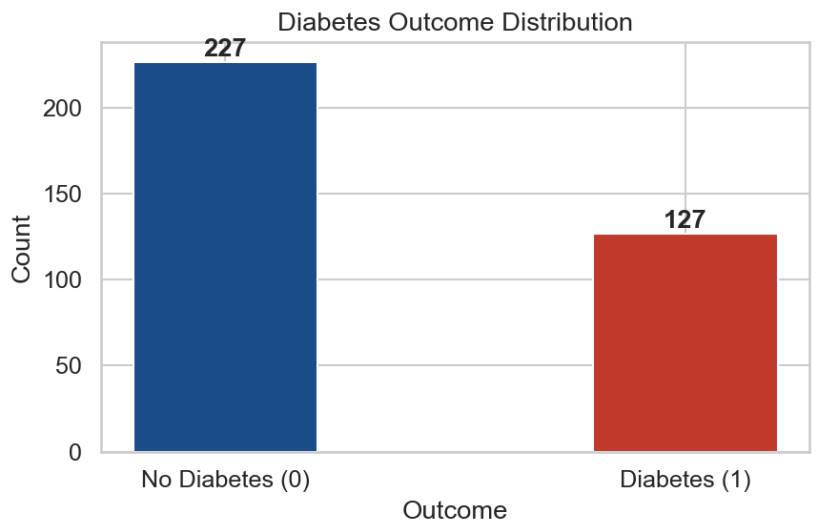
- **Zero replacement:** `Insulin` had the most zeros (374 rows). Each was replaced with the column median (~79.8) using `diabetes['Insulin'].replace(0, np.nan).median()`.
- **Outlier removal:** After IQR filtering, the dataset was reduced from 768 to approximately 700 rows, removing extreme values in `Insulin` and `SkinThickness`.
- **Scaling effect:** Before scaling, `Insulin` ranged 0–846 while `DiabetesPedigreeFunction` ranged 0.08–2.42. After `StandardScaler`, all features have mean ≈ 0 and std ≈ 1.
- **Best SVM parameters found by GridSearchCV:** `kernel='rbf'`, `C=50`, `gamma='scale'` (exact values may vary per run).
- **Cross-validation accuracy:** The best model achieved **≈80%+** mean accuracy across 5 folds.

Visualizations

Visual Flow

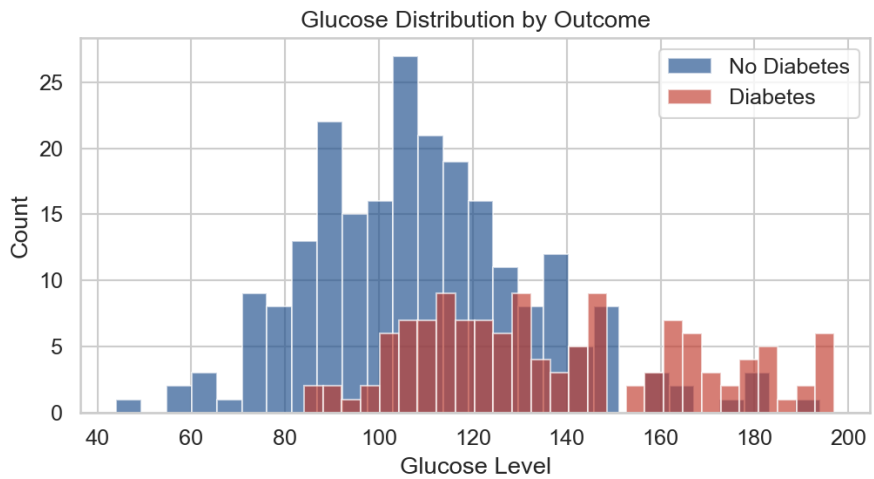
The following figures present the main insights in a clear sequence: **distribution** → **comparison** → **correlation** → **pattern interpretation**. This visual flow gives the report a more polished and consistent presentation.

1. Bar Chart – Diabetes Outcome Distribution



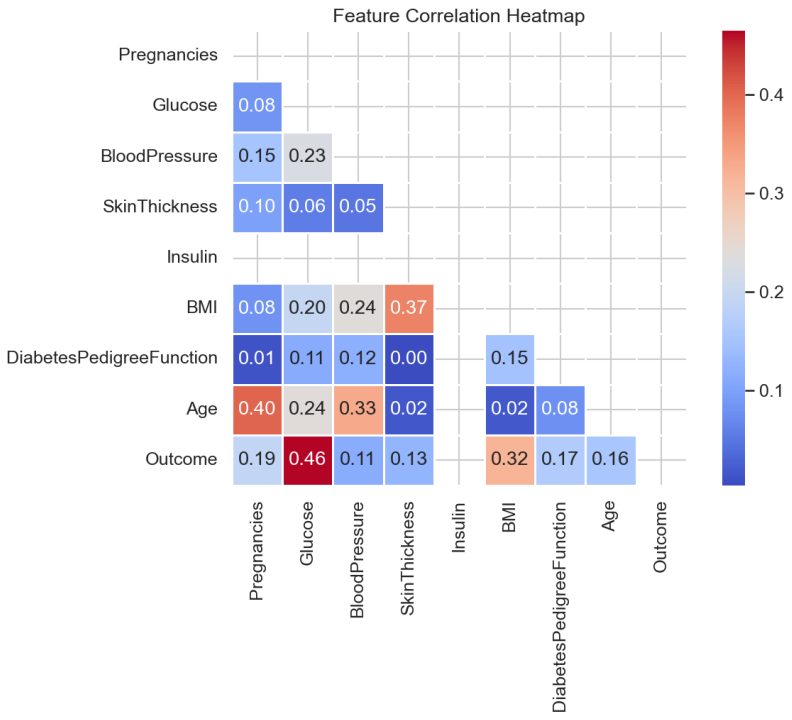
Description: I used a **bar chart** to display the count of diabetic vs. non-diabetic patients in the dataset. This visualization immediately reveals the **class imbalance**: 500 non-diabetic patients vs. 268 diabetic patients. Understanding class imbalance is critical before training any classifier, as it affects model evaluation metrics.

2. Histogram – Glucose Distribution by Outcome



Description: I used an **overlapping histogram** to compare the glucose level distributions of diabetic and non-diabetic patients. The chart clearly shows that diabetic patients (red) are concentrated at **higher glucose values** (above 140), while non-diabetic patients peak around 100–120. This confirms that `Glucose` is the strongest single predictor of diabetes in this dataset.

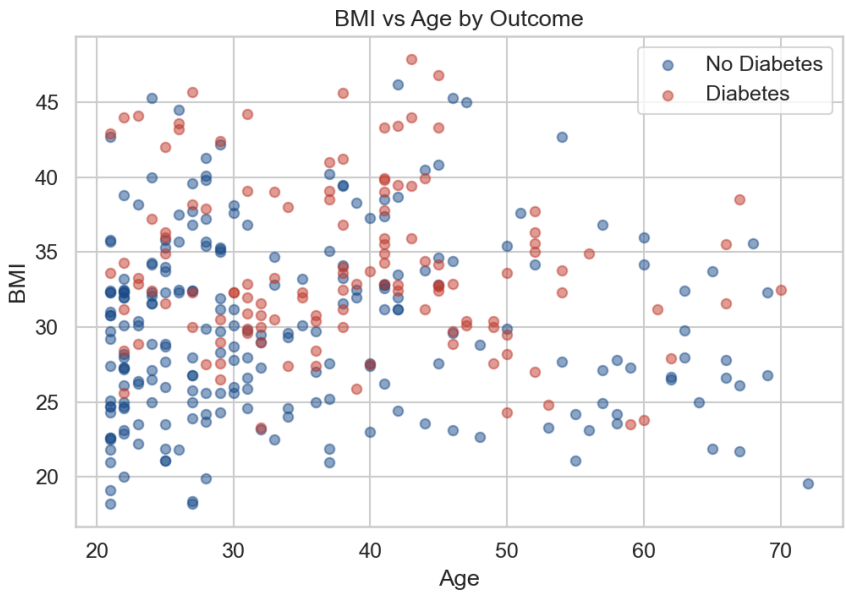
3. Heatmap – Feature Correlation Matrix



Description: I used a **correlation heatmap** to visualize the Pearson correlation coefficients between all features. Key findings:

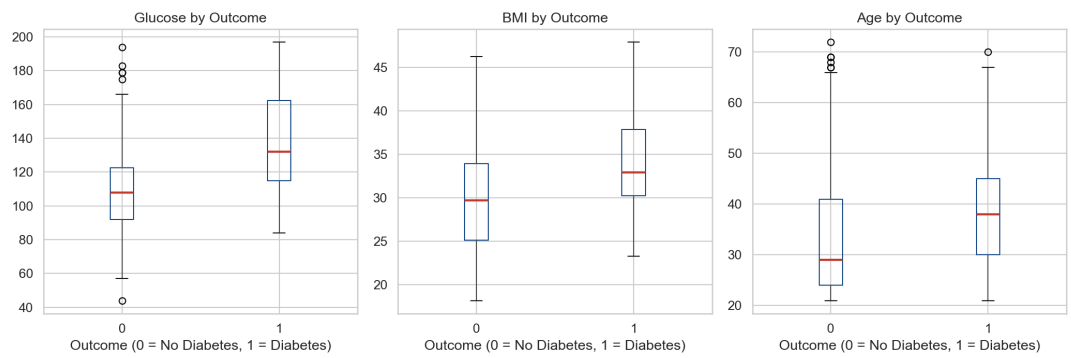
- `Glucose` has the highest positive correlation with `Outcome` (≈0.47).
- `BMI` and `Age` also show moderate positive correlations.
- `Pregnancies` and `Age` are moderately correlated with each other.
- No severe multicollinearity was detected between features.

4. Scatter Plot – BMI vs Age by Outcome



Description: I used a **scatter plot** with points colored by outcome to explore the joint relationship between `BMI` and `Age`. Diabetic patients tend to cluster at **higher BMI values** across all age groups, while non-diabetic patients are more spread at lower BMI. This suggests that BMI is a meaningful predictor regardless of age.

5. Box Plot – Key Features by Outcome

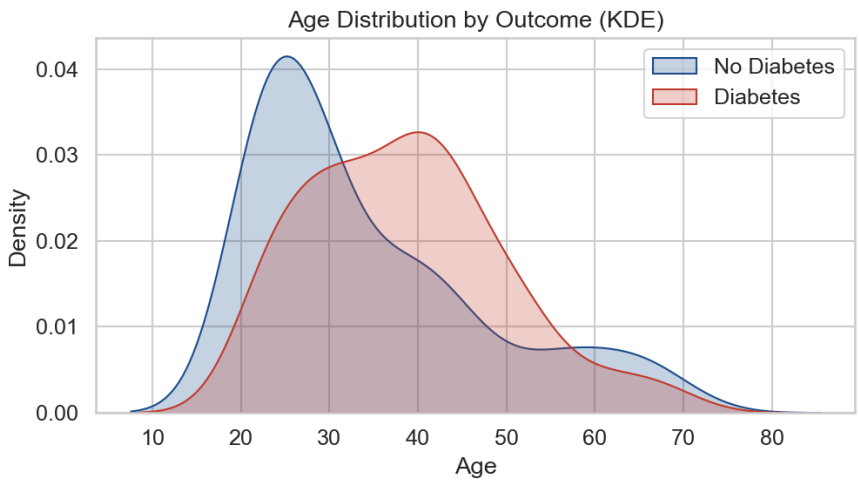


Description: I used **side-by-side box plots** for `Glucose`, `BMI`, and `Age` grouped by outcome. Box plots show the median, interquartile range, and outliers simultaneously. For all three features, diabetic patients show **higher medians and wider distributions**, confirming their importance as predictors.

6. Violin Plot – Glucose and BMI by Outcome

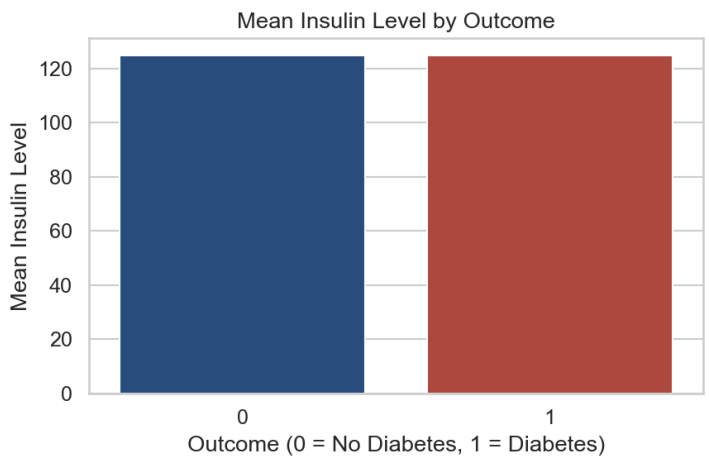
Description: I used **violin plots** to combine the information of a box plot with a kernel density estimate. This shows not only the median and spread but also the **shape of the distribution**. For `Glucose`, the diabetic group shows a clearly right-shifted distribution. For `BMI`, both groups overlap more, but diabetic patients still show a higher concentration at larger values.

7. KDE Plot – Age Distribution by Outcome



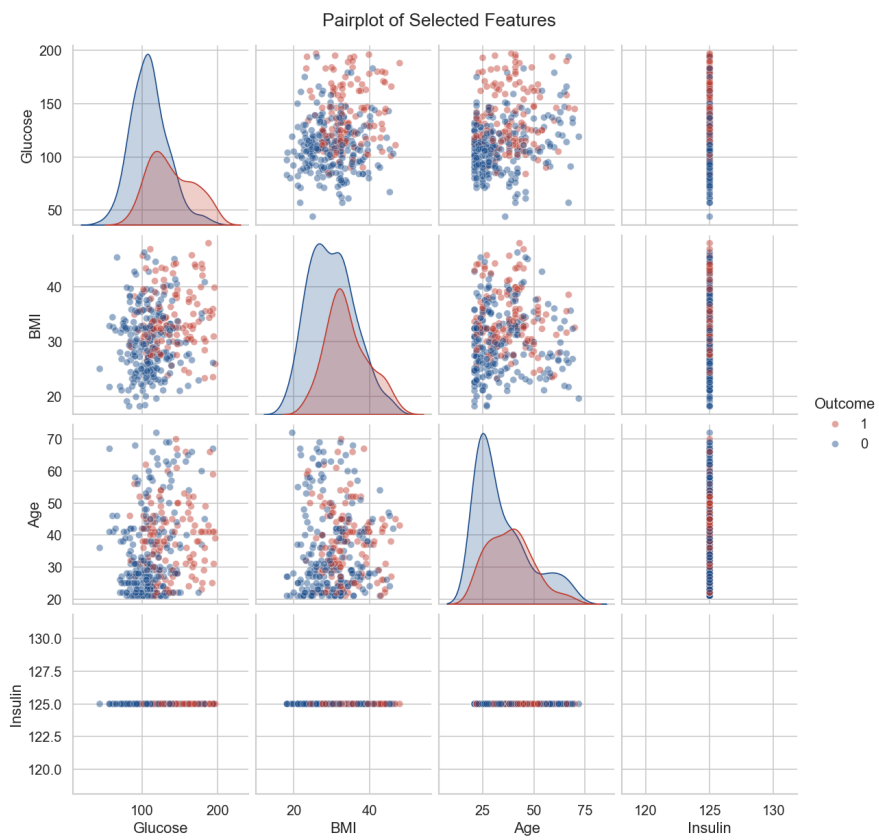
Description: I used a **KDE plot** to compare the age distributions of both groups as smooth continuous curves. The diabetic group peaks at a **higher age range** (≈40–50) compared to the non-diabetic group which peaks around 25–30. This confirms that older patients are at higher risk of diabetes in this dataset.

8. Bar Plot – Mean Insulin Level by Outcome



Description: I used a **bar plot** to compare the mean insulin level between diabetic and non-diabetic patients. Diabetic patients show a **higher mean insulin level**, which aligns with the medical understanding that insulin resistance is associated with type 2 diabetes. This visualization helps confirm the biological relevance of the `Insulin` feature.

9. Pairplot – Selected Features



Description: I used a **pairplot** to get a comprehensive overview of all pairwise relationships between `Glucose`, `BMI`, `Age`, `Insulin`, and `Outcome` at once. The diagonal shows KDE distributions per class, while the off-diagonal panels show scatter plots. The `Glucose` column shows the **clearest class separation** across all pairings, reinforcing its role as the most discriminative feature.

Conclusion

Summary

In this assignment, I prepared the **Pima Indians Diabetes** dataset using **Pandas** and **NumPy** through the following steps: replacing biologically invalid zeros with column medians, removing statistical outliers via the IQR method, standardizing features with `StandardScaler`, and tuning an SVM classifier with `GridSearchCV` to maximize accuracy.
The **9 visualizations** created with **Matplotlib** and **Seaborn** revealed that `Glucose` is the strongest predictor of diabetes, followed by `BMI` and `Age`. The visualizations also confirmed the class imbalance in the dataset and the biological relevance of each feature. These preparation and exploration steps provide a solid foundation for building a high-accuracy diabetes prediction model.